

METASPLOIT PENETRATION TEST REPORT

Metasploitable2 — Full Exploitation Assessment

Prepared by: Ekene Henry Uwaezuoke

Role: SOC Analyst & Penetration Tester

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Classification: CONFIDENTIAL

1. Executive Summary

A comprehensive penetration test was conducted against Metasploitable2 using Metasploit Framework v6.3.43. The assessment identified and exploited multiple critical vulnerabilities resulting in full root-level access to the target system via three separate attack vectors. All findings are documented with CVSS scores, MITRE ATT&CK; mappings, and remediation recommendations.

2. Scope & Methodology

Item	Details
Target System	Metasploitable2 (Deliberately Vulnerable VM)
Target IP	192.168.56.101 [LAB ENVIRONMENT]
Attacker IP	192.168.56.1 (Kali Linux)
Testing Type	Black Box Network Penetration Test
Tools Used	Nmap, Metasploit Framework, Netcat
Test Date	2026-06-26
Tester	Ekene Henry Uwaezuoke

3. Services Discovered

Port	Service	Version	Risk
21/tcp	FTP	vsftpd 2.3.4	CRITICAL

22/tcp	SSH	OpenSSH 4.7p1	HIGH
23/tcp	Telnet	Linux telnetd	HIGH
80/tcp	HTTP	Apache 2.2.8	MEDIUM
139/445	Samba	3.X-4.X	CRITICAL
1524/tcp	Bindshell	Root Shell	CRITICAL
3306/tcp	MySQL	5.0.51a	HIGH
5432/tcp	PostgreSQL	8.3.0	HIGH
5900/tcp	VNC	Protocol 3.3	HIGH
6667/tcp	IRC	UnrealIRCd	HIGH
8180/tcp	HTTP	Apache Tomcat	MEDIUM

4. Findings Summary

#	Vulnerability	Severity	CVSS	MITRE ATT&CK
F-01	vsftpd 2.3.4 Backdoor RCE	CRITICAL	10.0	T1190
F-02	Samba Usermap Script RCE	CRITICAL	10.0	T1210
F-03	Bindshell Root Access Port 1524	CRITICAL	10.0	T1049
F-04	SSH Weak Credentials	HIGH	8.8	T1110
F-05	Telnet Unencrypted Service	HIGH	7.5	T1557
F-06	MySQL Exposed Port 3306	HIGH	7.5	T1190
F-07	VNC Exposed Port 5900	HIGH	7.5	T1021
F-08	Apache 2.2.8 Outdated	MEDIUM	5.3	T1190

5. Detailed Findings

F-01: vsftpd 2.3.4 Backdoor RCE

Description: vsftpd 2.3.4 contains a backdoor triggered by a smiley face ':' in the username field opening a shell on port 6200.

Module: exploit/unix/ftp/vsftpd_234_backdoor

Result: Root shell obtained on target system.

CVSS Score: 10.0 (CRITICAL)

MITRE ATT&CK;: T1190 — Exploit Public-Facing Application

Impact: Full root access to target system.

Remediation: Upgrade vsftpd immediately. Remove backdoored version.

F-02: Samba Usermap Script RCE

Description: Samba 3.X contains a vulnerability in the username map script allowing command injection.

Module: exploit/multi/samba/usermap_script

Result: Command shell session opened. Root access confirmed.

CVSS Score: 10.0 (CRITICAL)

MITRE ATT&CK:: T1210 — Exploitation of Remote Services

Impact: Full root access to target system.

Remediation: Update Samba to latest version. Disable unnecessary shares.

F-03: Bindshell Root Access — Port 1524

Description: Port 1524 is running a bindshell providing direct root access with no authentication required.

Command: nc 192.168.56.101 1524

Result: Immediate root shell. whoami returned 'root'.

CVSS Score: 10.0 (CRITICAL)

MITRE ATT&CK:: T1049 — System Network Connections Discovery

Impact: Any attacker on the network has instant root access.

Remediation: Close port 1524 immediately. Audit all listening ports.

F-04: SSH Weak Default Credentials

Description: SSH service accepts default credentials msfadmin/msfadmin.

Module: auxiliary/scanner/ssh/ssh_login

Result: Successful SSH login with default credentials.

CVSS Score: 8.8 (HIGH)

MITRE ATT&CK:: T1110 — Brute Force

Impact: Unauthorized remote access to system.

Remediation: Change default credentials. Implement SSH key authentication. Disable password auth.

6. Conclusion

The penetration test against Metasploitable2 revealed multiple critical vulnerabilities allowing complete system compromise via three separate attack vectors. The most severe finding was the vsftpd 2.3.4 backdoor and the exposed bindshell on port 1524 which provide instant root access with no authentication. Immediate remediation is required for all critical findings. All findings have been documented with CVSS scores, MITRE ATT&CK; mappings, and actionable remediation guidance.

Report prepared by: Ekene Henry Uwaezuoke

SOC Analyst & Penetration Tester

Portfolio: bihghefty.github.io/cyberportfolio